Proposition Logic

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function TT-ENTILES? (KB, α) returns true or false
inputs KB, the knowledge base, a sentence in propositio
α, the query, a sentence in proposition logic
symbols ← a list of the preposition symbols in KB
return TT-CHECK-ALL (KB, α, symbols, {}) and α

function TT-CHECK-ALL (KB, α, symbols, models) returns
true or false
if empty? (symbols) then
if PL-TRUE? (KB, model) then returns PL-TRUE?
(α, model)
else
returns TRUE
else do
p ← FIRST
rest ← REST
return (TT-CHECK-ALL (KB, α, rest, model ∪ {P=true})
and TT-CHECK-ALL (KB, α, rest, model ∪ {P=false})

Consider a knowledge base KB that contains the following propositional logic.

$Q \rightarrow P$

$P \rightarrow \neg Q$

$Q \vee R$

(i) Construct a truth table that shows the truth valu
of each sentence in KB and indicate the models in
which the KB is true

$KB = (Q \rightarrow P) \wedge (P \rightarrow \neg Q) \wedge (Q \vee R)$

P	Q	R	¬Q	Q→P	P→¬Q	QVR	KB	P→P	Q→R
T	T	T	F	T	F	T	F	T	T
T	T	F	F	T	F	T	F	T	F
T	F	T	T	T	T	T	T	T	T
T	F	F	T	T	T	F	F	T	T
F	T	T	F	F	T	T	F	F	T
F	T	F	F	F	T	T	F	T	F
F	F	T	T	T	T	T	T	F	T
F	F	F	T	T	T	F	F	T	T

(ii) Does KB entail R? $\alpha = R$.

True.

KB entails R

→ i.e, $KB \models R$

(iii) Does KB entails $R \to P$?

Here $\alpha = R \to P$

False.

KB does not entails $R \to P$ (in 7th row)

(iv) Does KB entail $Q \to R$?

Here $\alpha = Q \to R$

~~False~~

True. KB entails $Q \to R$?

i.e, $KB \models (Q \to R)$.